



Feed the Soil. Power the Future.

Smart Fertilizer System

SEABI TECH INNOVATIONS

Farming problems

- Most famers in South Africa face problems such as:
 - poor plant growth
 - uneven nutrient distribution
 - fertilizer waste
 - high labor costs, and
 - environmental pollution.
- These leads to:
 - loss of income due to plant health
- Research from the University of Pretoria indicates that many Famers in Rural Areas and in South African are already experiencing soil degradation, nutrient imbalances, water quality impacts, and declining sustainability after decades of cultivation.





PROBLEM SOLUTION

The smart fertilizer system:

- Reads the soil condition
- Test the soil.
- Analysis and give recommended fertilizers and crops.
- Monitor temperature in Greenhouse



Competitors

Aerobotics

khula!

MEANS TO GROW



SMART FERTILIZER
Seabi Tech Innovations

Good afternoon, April

Seabi Green Farm

Field 01 · Maize

SMART-FERT-001 · Offline · demo data shown DEMO



CACHED

Soil Condition

67%

Healthy range



CACHED

Temperature

24°C

Good for growth

PHONE APP → CLOUD → DEVICE

Prototype pathway



PWA



Seabi cloud



ESP32



Soil health: **GOOD**



Fertilizer needed: **NO**

Sensors are reporting moisture, climate and pump state.

Seabi Tech Innovations · Smart Fertilizer System · Last known data shown safely



Dashboard



Soil



History

What Makes My Innovation Different?

- Uses real-time soil monitoring.
- Automatically dispenses fertilizer when needed.
- Reduces human error and guesswork.
- GrowPilot App (Device App)
- Combines monitoring and dispensing in one system.
- Low-cost compared to many smart farming solutions.
- Easy to use and maintain.
- Helps farmers save money on fertilizer.
- Supports precision agriculture at an affordable cost.

Target Market/Customers

- Small-scale farmers.
- Emerging farmers.
- Rural farming communities.
- Community gardens.
- Agricultural training centers.
- Schools with agricultural projects.
- Sustainable farming initiatives.
- Farmers in need of modern farming technology.



Business Revenue

ITEM	COST PRICE	SELLING PRICE	PROFIT
Production of the smart fertilizer system	R5 500	R625 p/m (no contract) + subscription	R2 000 annual
Maintenance	Travelling costs	Travelling costs + R500	R500
Repair	Item costs + travelling costs	Item costs + travelling costs + R600	R600

Waste Management

TYPE OF WASTE	MANAGEMENT SYSTEM
Cut-offs during designing	Recycled
Damaged cables	Recycled
Dead batteries	Exchanged
Damaged electronic parts	Recycled

Environmental Benefits



Reduces fertilizer
waste



Supports sustainable
agriculture



Improves productivity



THANK YOU.

Let us grow future together

